

Homework 2 - Distributed Remote A2A System

Repository

github.com/Sudashiii/dibse_industrial_computing

One-page implementation and verification summary

Implementation

- Registry with capability-based lookup
- Search, Evidence and Synthesis agents
- A2A JSON-RPC message/send over HTTP
- Docker Compose with health-gated startup

Verification

- 3 registered agents per run
- 3 sample requests completed
- 5 search hits -> 5 evidence items
- LiteLLM liveness and Compose passed

Relevant execution output

A2A workflow - focused execution evidence

```
[2026-09-18T11:52:48+00:00] COMMAND docker compose up --build --detach --wait litellm registry search-agent evidenc...
LITELLM_HEALTH status=ok endpoint=/health/liveliness
[2026-09-18T11:53:09+00:00] COMMAND docker compose --profile workflow run --rm orchestrator --question Welche Vorte...
INFO REGISTRY_LOOKUP capability=research.search
INFO A2A_CALL step=search agent_id=search-agent
INFO REGISTRY_LOOKUP capability=research.extract_evidence
INFO A2A_CALL step=evidence agent_id=evidence-agent
INFO REGISTRY_LOOKUP capability=research.synthesize
INFO A2A_CALL step=synthesis agent_id=synthesis-agent
INFO WORKFLOW_COMPLETED question='Welche Vorteile und Grenzen haben Retrieval-Systeme?' agents=search-agent,evidenc...
SUMMARY agents=3 search_hits=2 evidence_items=2 synthesis_findings=2
INFO WORKFLOW_COMPLETED question='Wie wichtig sind Evaluation und Zitate für RAG-Systeme?' agents=search-agent,evid...
SUMMARY agents=3 search_hits=1 evidence_items=1 synthesis_findings=1
INFO WORKFLOW_COMPLETED question='Welche risk-, compliance- und privacy-Aspekte gelten für high stakes organization...'
SUMMARY agents=3 search_hits=2 evidence_items=2 synthesis_findings=2
CLEANUP compose_down=ok
```

Only registry lookups, agent calls and workflow totals are shown.

Sample evidence

A2A sample requests - parsed workflow results

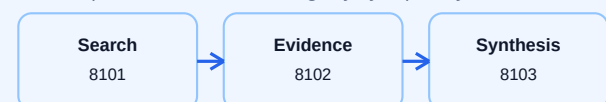
EXAMPLE 1 QUESTION: Welche Vorteile und Grenzen haben Retrieval-Systeme? REGISTRY -> 3 agents SEARCH Hits: P-001, P-002 EVIDENCE Items: 2 SYNTHESIS Findings: 2 FIRST FINDING: [P-001] Retrieval-augmented generation can ground generated answers in a changing knowledge base and expose supporting	EXAMPLE 2 QUESTION: Wie wichtig sind Evaluation und Zitate für RAG-Systeme? REGISTRY -> 3 agents SEARCH Hits: P-003 EVIDENCE Items: 1 SYNTHESIS Findings: 1 FIRST FINDING: [P-003] Evaluation should separate retrieval quality, answer correctness and citation completeness; a high retrieval	EXAMPLE 3 QUESTION: Welche risk-, compliance- und privacy-Aspekte gelten für high stakes organization search? REGISTRY -> 3 agents SEARCH Hits: P-004, P-005 EVIDENCE Items: 2 SYNTHESIS Findings: 2 FIRST FINDING: [P-004] Human review is useful for high-stakes research decisions because automated systems may miss ambiguous evidence or outdated
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All values are parsed from workflow samples (only no confidentiality or external policy rules are shown).

Three live questions; full JSON and log are linked below.

Integration flow

Each step is selected from the Registry by capability.



Registry: port 8000 | Orchestrator: one-shot Compose profile